

Washing Machine Simulator

A software design of a washing machine has the following functional requirements.

Present the overall design and write a simulation program to achieve the required functionality. Assume that you have all the required sensors, actuators and interfacing circuitry available. The simulation program should support the below-mentioned functionality.

1. Build the interface for feeding simulation inputs, for example, timer dial, push buttons, door status, detergent status, and power status.
2. Build an output interface showing the current status of the washing machine, including the selected mode, machine state, remaining time, door status, and detergent status.
3. The machine should support three modes of operation:
 - Heavy – 45 minutes
 - Normal – 30 minutes
 - Light – 20 minutes
4. The machine must not start without a valid wash mode being selected.
5. The machine must not start if the door is open. The door must be closed before the washing cycle can begin.
6. The machine must not start if the detergent compartment is empty. If the “Start” button is pressed while the detergent compartment is empty, the start request should remain pending. As soon as detergent is filled, the machine should automatically start without requiring another Start input, provided that a valid mode has been selected and the door is closed.
7. When washing is in progress, the door must remain locked. The selected wash mode must not be changed while the machine is running.
8. If the “Start” button is pressed while the machine is already running, the input should have no effect and the current washing cycle should continue normally.
9. The machine should have an emergency “Abort” button. If “Abort” is pressed while washing is in progress, the washing cycle should stop immediately, and the door should be unlocked.
10. When the washing timer reaches zero, the washing cycle should be considered complete. The machine should stop the washing operation and unlock the door.
11. In case of a power failure, the machine should preserve its current state and the remaining time of the washing cycle. When power is restored, the machine should resume the unfinished cycle from the point at which it stopped, without restarting the completed portion of the cycle.
12. The simulation should handle invalid user inputs appropriately. Invalid inputs must not modify the current state of the washing machine, and an appropriate message should be displayed to the user.

Simulation Requirements

The program should provide a command-line interface through which the user can simulate the required inputs and observe the machine's behaviour.

The implementation should be organised into multiple C source (.c) and header (.h) files, with different functional components separated into appropriate modules.